

Zhenyuan Zhang

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EDUCATION

PhD candidate in Human Movement Biomechanics

Jul 2023 – Present

Research Institute for Sport and Exercise Science
Liverpool John Moores University, United Kingdom

Master of Science in Sport and Clinical Biomechanics

Sep 2020 – Nov 2021

School of Sport and Exercise Science
Liverpool John Moores University, United Kingdom

Bachelor of Education in Human Movement Science

Sep 2016 – Jun 2020

School of Exercise and Health Science
Chengdu Sport University, China

SKILLS

Languages: Python, Matlab, Git, Shell, L^AT_EX

Computer Simulations: OpenSim, Visual3D, Pyomeca, MyoSuite

Biomechanics Tools: Optical Motion Capture, Inertial Measurement Units, Force Plates, Electromyography

High-Performance Computing: Slurm, AWS, Dask

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn

TECHNICALITIES

Machine Learning and Biomechanics Researcher

Jul 2023 – Present

SportScientia Ltd., Industry Partnership
Remote

- Developed and validated deep learning neural network models to estimate ground reaction forces from smart insoles instrumented with pressure sensors and inertial measurement units for various movement tasks.
- Validated smart insoles with customized sensor fusion algorithm for measuring spatiotemporal gait parameters against optical motion capture and force plates for athletic performance and load monitoring applications.
- Assisted in developing cloud computing pipelines with AWS to automate data processing and analysis for the smart insole system.

Biomechanics Laboratory Technician

Sep 2025 – Jan 2026

Liverpool John Moores University, Contracted
United Kingdom

- Built up and fine-tuned 3 advanced biomechanics systems with the senior technician, provided professional training for staff and students. I am also responsible for managing them for both teaching and research activities.
- System1: 10 Qualysis[®] Arqus motion capture cameras integrated with 8 Qualysis[®] Miquis cameras for marker-less motion capture, 2 Kistler[®] force plates, 16 Delsys[®] Trigno EMGs.
- System2: 8 Qualysis[®] Arqus motion capture cameras integrated with a Treadmetrix[®] treadmill (AMTI[®] force plate embedded), 16 Delsys[®] Trigno EMGs and 8 Noraxon[®] EMGs.
- System3: 14 Vicon[®] Vero motion capture cameras integrated with 16 Vicon[®] T-series cameras, 2 Kistler[®] force plates, 8 Vicon[®] Blue Trident IMUs and 16 Delsys[®] Trigno EMGs.

Graduate Research Assistant

Nov 2021 – Jul 2023

Liverpool John Moores University, Contracted
United Kingdom

- Assisted in commercial projects with New Balance Athletics, USA to test biomechanical interactions between soccer boots with different studs and artificial turfs using high-speed motion capture and force plates.
- Assisted in commercial projects with New Balance Athletics, USA to test effects of different running shoes on lower limb biomechanics and muscle co-contractions during treadmill running using motion capture integrated with instrumented treadmill and EMGs.

PUBLICATIONS

- Zhang, Z.**, Verhuel, J., Robinson, M., and Lake, M. Estimating ground reaction forces in dynamic sports movements using instrumented insoles and deep learning. *Oral Presentation at XXX Congress of International Society of Biomechanics*, p.85033, Stockholm, Sweden. (2025)
- Yang, C., Yang, Y., Xu, Y., **Zhang, Z.**, Lake, M., and Fu, W. Whole leg compression garments influence lower limb kinematics and associated muscle synergies during running. *Frontiers in Bioengineering and Biotechnology*, 12, 1310464. (2024)
- Zhang, Z.** and Lake, M. Rate of knee flexion at the instant of landing during running can influence initial knee joint stiffness estimates due to running shoe cushioning. *Oral Presentation at XXIX Congress of International Society of Biomechanics*, p.314, Fukuoka, Japan. (2023)
- Zhang, Z.** and Lake, M. A re-examination of the measurement of foot strike mechanics during running: the immediate effect of footwear midsole thickness. *Frontiers in Sports and Active Living*, 4, 824183. (2022)
- Zhang, Z.** and Lake, M. A comparison of unmatched and matched filtering approaches for knee joint stiffness calculation during running. *Oral Presentation at 40th International Society of Biomechanics in Sports, Proceedings Archive*, 40(1), 807, Liverpool, United Kingdom. (2022)

PROJECTS

TechLayer | [GitHub \(Coming Soon\)](#)

- A Python project which trained and implemented a deep learning model to predict Ground Reaction Forces (GRFs) from IMU and pressure sensor data collected from instrumented insoles during various dynamic sports movements.
- The model is trained and validated on a dataset of 32 participants using High-Performance Computing (HPC) clusters and demonstrated high accuracy in predicting vertical and anterior-posterior GRFs against force plates.

Wearable_IK | [GitHub \(Coming Soon\)](#)

- A Python project integrating open-source sensor fusion algorithms and OpenSim API to estimate joint kinematics from IMU data and validating the results against optical motion capture data.
- It automates the entire workflow from data loading, preprocessing, sensor fusion, sensor-to-segment calibration, inverse kinematics, to results visualization
- It also features parallelized computing to speed up processing and a calibration-free approach for IMU sensors as it does not require magnetometers.

Wearable_System | [GitHub \(Under Development\)](#)

- A Python project integrating **TechLayer** and **Wearable_IK** to generate muscle-driven simulations of human movement while solving optimal control problems from wearable sensor (instrumented insoles + IMUs) using OpenSim Moco. It also aims to validate the simulation results against traditional biomechanics measurement system using optical motion capture, force plates and EMGs.

My_Website | [GitHub](#)

- A project to build and deploy my personal academic website from an open-source TypeScript template for fun.

TEACHING

Graduate Teaching Assistant

Nov 2021 – Nov 2025

Liverpool John Moores University, Contracted
United Kingdom

- Assisted in biomechanics lectures and practical laboratory sessions for optical motion capture cameras, force plates, IMUs and EMGs at undergraduate and master levels.
- Provided one-on-one tutorial support for students' projects and technical training of setting up and using the equipment for data collection of students' projects.

SCHOLARSHIPS

University-Industry Matched PhD Fund: A three-year matched funding package awarded jointly by Liverpool John Moores University and industry partner to support collaborative PhD research. (Jul 2023)

LJMU International Achievement Scholarship: A scholarship awarded to international students with excellent academic performance to pursue postgraduate studies at Liverpool John Moores University. (Sep 2020)

PUBLIC SERVICES

IBS 2025 Congress Session Chair: Served as the chair for parallel session "Wearable Measurement Devices I" at the XXX Congress of the International Society of Biomechanics (Stockholm, Sweden). (2025)

REFERENCES

References available upon request.